

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 12

Information Theory and the Cost of Description

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. You discover that a very simple model (few parameters) fits your data almost as well as a very complex model (many parameters). What does this tell you about the entropy and structure of the data-generating process?

2. What is the information-theoretic interpretation of overfitting?

3. A colleague says: “Mutual information is just a better version of correlation.” What is right about this? What is missing?

4. You are given a perfectly compressed file (compressed to its entropy limit). Can you learn anything about the structure of the original data from the compressed file? Why or why not?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
